

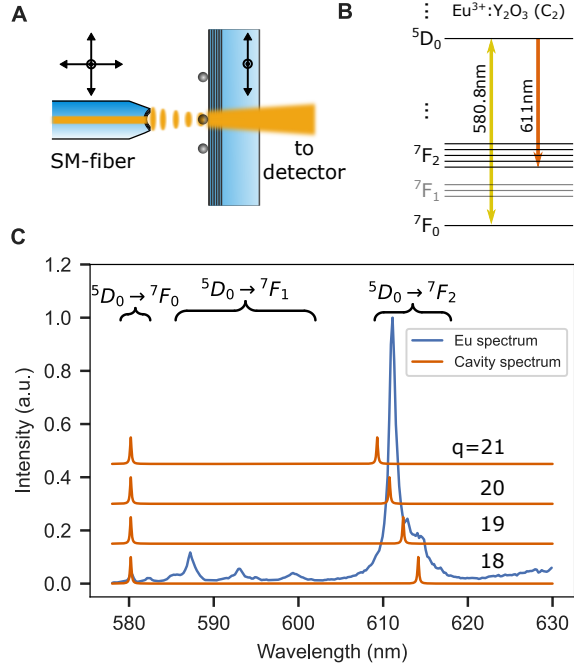


Fig. 1: **A:** schematic drawing of the fiber-based Fabry-Pérot microcavity. Laser light enters the cavity via a single mode optical fiber (SM-fiber), and the transmission is collected by a photodiode or a single photon counting module (see optical setup in supplementary material). **B:** Relevant level scheme of europium. **C:** emission spectrum of the ⁵D₀ excited state for the transitions depicted in B (blue line), together with a simulation of the cavity spectrum for longitudinal mode orders $q = 18 - 21$ of the 580 nm resonance (red lines).

optically excited state. This renders single ion read-out difficult due to low fluorescence count rates. However, enhancing the density of states of electromagnetic modes and increasing their vacuum field by an optical cavity gives rise to the Purcell effect [8], which manifests as a lifetime shortening. The Purcell factor F_P , quantifying the increase in the spontaneous emission rate, for a Fabry-Pérot type cavity is given as:

$$F_P = \frac{6}{\pi^3} \left(\frac{\lambda}{n} \right)^2 \frac{\mathcal{F}}{w_0^2}. \quad (1)$$

Here, $\mathcal{F}$ denotes the cavity finesse, w_0 is the mode waist of the Gaussian cavity mode, λ the transition wavelength, and n the refractive index. Using nano- and micro-scale optical cavities, single REI readout has been demonstrated for various species in the recent years, but not for Eu³⁺ to date [9–14].

Here, we present important steps towards the detection of single europium ions. We study the optical and coherence properties of small ion ensembles inside individual europium-doped nanoparticles (NPs)

at cryogenic temperatures. To reach a high Purcell factor, we incorporate the NPs into a fiber-based Fabry-Pérot microcavity [15, 16], which features a high finesse and a small mode waist. The cavity’s spatial tunability allows us to study six different NPs and to perform cavity-enhanced spectroscopy. The spectral tunability of the cavity is then employed to demonstrate a multimodal Purcell-enhancement of two decay channels of the same excited state. Furthermore, we use transient spectral hole burning to determine an upper bound of the homogeneous linewidth. The measured parameters allow for a realistic estimation of the count rate that we expect from a single europium ion. The system performance demonstrated here should enable the detection and characterization of single ions.

2 Methods

We operate a fully-tunable, open-access, fiber-based Fabry-Pérot microcavity inside a liquid helium flow cryostat. The cavity assembly is mounted on the cold plate of the cryostat and is cooled by helium exchange gas down to a minimum temperature of 3.5 K. The microcavity consists of a concave fiber mirror and a macroscopic planar mirror carrying the nanoparticles as shown in Fig. 1A. We use a single-mode (SM) optical fiber where a concave profile with radius of curvature of 25 μm has been machined onto the end facet by CO₂ laser machining [17, 18]. The fiber is then coated with a distributed Bragg reflector (DBR) mirror with 25 ppm transmission. The planar opponent is a DBR mirror with higher transmission of 200 ppm to optimize the out-coupled rate of photons R_{out} (see below) in the presence of mirror loss and nanoparticle scattering. Together with the absorption and scattering losses of the mirrors, this results in a bare cavity finesse of 17,500 at 580 nm and 9,500 at 611 nm. The mode waist at the planar mirror is about 1.4 μm , resulting in a nominal Purcell factor of $F_P = 580$ (330) at 580 nm (611 nm) according to Eq. 1. The fiber position can be fine tuned with sub-picometer precision in all directions by piezo actuators. Electrical motors achieve a coarse tuning of the cavity length over hundreds of microns and lateral movement of the mirror of up to one mm [19]. The nanopositioning stage is fully operational in the temperature range from room temperature down 3.5 K. The root mean square cavity length jitter, which has a strong influence on the Purcell-effect, can be as low as 2.5 pm under active stabilization [19]. For the mea-